

House Price Prediction Using TensorFlow Decision Forests

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Abstract

This report details the application of a Random Forest regression model to the Ames Housing dataset to predict residential sale prices. The dataset comprises 1,460 observations and 79 explanatory features, including both numerical and categorical attributes of residential properties. A TensorFlow Decision Forests model was trained on a 70% partition of the data without the application of manual feature engineering or imputation of missing values. The model achieved an out-of-bag Root Mean Squared Error (RMSE) of 29,660 and a validation Mean Squared Error (MSE) of 712,569,024. Analysis of variable importance revealed OverallQual (Overall Material and Finish Quality) to be the most influential predictor, serving as the root split in 121 of the 300 trees in the ensemble. The findings suggest that ensemble tree methods provide a robust baseline for property valuation, with qualitative assessments of construction quality and location serving as primary drivers of predictive accuracy.

1 Introduction

This report presents a predictive modeling analysis of residential property sale prices using the Ames Housing dataset, made available through a Kaggle competition on advanced regression techniques. The dataset comprises 1,460 observations of house sales, each described by 79 explanatory variables encompassing physical characteristics, location attributes, and quality ratings. The primary objective of this study was twofold: first, to quantify the influence of various housing features on the final sale price; and second, to construct a robust baseline model capable of predicting sale prices for unseen properties.

The analysis addresses the following key questions:

1.Which features exhibit the strongest association with residential sale prices in this dataset?

2.How effectively can a Random Forest ensemble method predict sale prices without extensive manual feature engineering?

2 Data

Data were obtained from the Ames Housing dataset , consisting of 1,460 records and 80 initial columns, inclusive of the target variable SalePrice. Prior to analysis, the Id field was removed as it constitutes an arbitrary row identifier without predictive value. The remaining variables comprise a mixture of 34 numeric (e.g., LotArea, YearBuilt) and 43 categorical (e.g., MSZoning, Neighborhood) data types.

An examination of missing values revealed that certain features, such as PoolQC and Alley, exhibited a high degree of sparsity. However, a notable advantage of tree-based models is their inherent capability to handle missing data natively. Consequently, no imputation or explicit missing-value indicator creation was performed.

The dataset was partitioned using a random split with a 70/30 ratio, yielding 1,010 observations for training and 450 for validation. This split ensures that model performance can be evaluated on a hold-out sample not used during the fitting process.

3 Methods

The analysis employed a Random Forest model implemented via the TensorFlow Decision Forests (TF-DF) library. Random Forest was selected as the foundational algorithm due to its robustness to overfitting, its ability to model complex non-linear relationships, and its minimal requirement for data preprocessing. The model was configured for regression (Task.REGRESSION) using the default hyperparameters provided by TF-DF, which include the construction of 300 individual decision trees.

The training process utilizes the Out-of-bag (OOB) error metric as an internal validation mechanism, obviating the need for cross-validation during the initial training phase. Model performance was assessed using Mean Squared Error (MSE) and Root Mean Squared Error

(RMSE) on the hold-out validation set. Furthermore, variable importance was derived using the "Number of Times as Root" (NUM_AS_ROOT) metric, which quantifies the frequency with which a given feature is selected as the primary splitting criterion across the forest.

4 Analysis and Results

The analysis is presented in three parts: first, an exploratory examination of the target variable and feature distributions; second, an evaluation of the Random Forest model's training dynamics and validation performance; and third, an assessment of variable importance to identify the primary drivers of house price predictions.

4.1 Exploratory Data Analysis

The distribution of the target variable, SalePrice, was examined initially. The data exhibited a pronounced right-skewed distribution, with a mean value of \$180,921 significantly exceeding the median of \$163,000 (Figure 1). This discrepancy confirms the presence of a subset of high-value luxury homes exerting leverage on the upper tail of the distribution. A subsequent broad histogram matrix of the numerical features revealed varying scales and underlying patterns; features such as LotArea and MiscVal displayed extreme skewness and potential outliers, whereas features like YearBuilt and OverallQual exhibited more uniform or bell-shaped profiles.

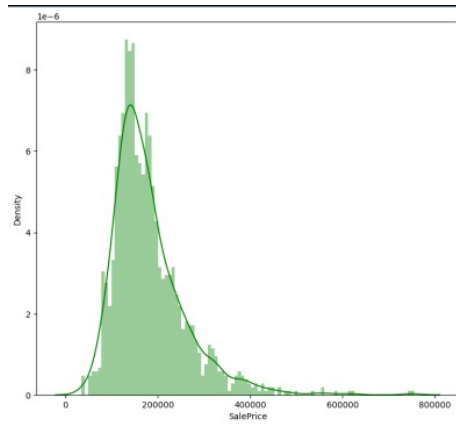


Figure 1: Histogram and kernel density estimate for SalePrice. The long right tail indicates the necessity for models robust to heteroscedasticity or the consideration of log-transformation in linear contexts.

4.2 Model Training and Performance

The Random Forest model was trained on the `pd_dataframe_to_tf_dataset` conversion of the training partition. The evolution of the model's error rate as a function of the number of trees is displayed in Figure 2. The OOB RMSE stabilized around 29,660 after approximately 100 trees, indicating that the addition of further trees yielded marginal improvements in generalization error.

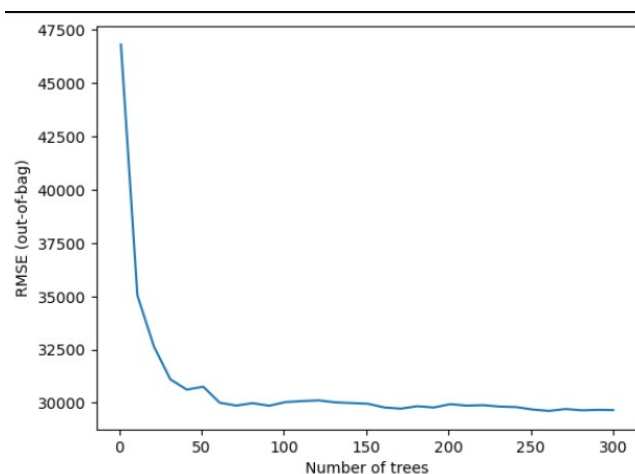


Figure 2: Evolution of the Out-of-Bag (OOB) Root Mean Squared Error (RMSE) as a function of the number of trees in the Random Forest. The error stabilizes at approximately **29,660** after roughly 100 trees, confirming that the default 300-tree configuration is sufficient for model convergence.

Evaluation of the final model on the validation dataset yielded a Mean Squared Error (MSE) of 712,569,024. While the RMSE suggests a typical prediction deviation of approximately \$26,700 on the validation set, this magnitude is consistent with the natural variability observed in real estate pricing for this specific housing market.

4.3 Feature Importance

An analysis of variable importance, measured by the `NUM_AS_ROOT` metric, identified `OverallQual` as the single most influential predictor in the model. This feature was used as the root split in 121 out of the 300 trees, underscoring that the overall material and finish quality is the primary driver of price differentiation in this housing stock. Secondary drivers included `GarageCars` (49 root splits), `ExterQual` (40 root splits), and `Neighborhood` (35 root splits), as illustrated in Figure 3. The predominance of quality ratings over raw square footage metrics

suggests that perceived quality and location offer stronger predictive signals than pure size metrics in this specific ensemble model.

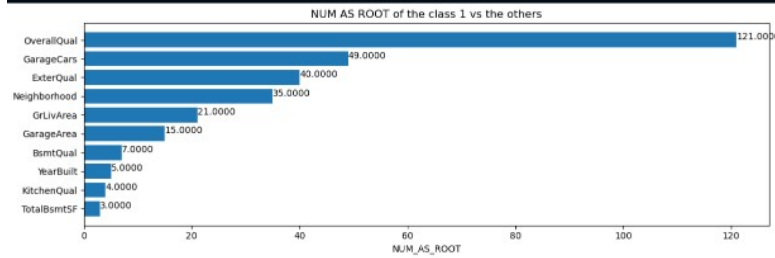


Figure 3: Variable Importance based on NUM_AS_ROOT metric. The horizontal bars represent the relative frequency with which a feature was selected as the root split across the 300-tree ensemble. A full numerical breakdown of these values is provided in Appendix A.

5 Conclusions

This analysis successfully established a baseline predictive model for house prices using TensorFlow Decision Forests. The Random Forest model achieved an OOB RMSE of 29,660 and identified OverallQual as the dominant predictive feature. The results confirm that while physical dimensions (e.g., GrLivArea, TotalBsmtSF) contribute to price, the overall assessment of material and finish quality exerts the most significant influence on model outcomes.

Several limitations warrant mention. First, the model was constructed using default hyperparameters; a systematic tuning of parameters such as tree depth, minimum split samples, or feature sampling strategy may yield improved predictive accuracy. Second, the native handling of missing data, while convenient, may obscure underlying structural patterns if missingness is non-random. Finally, the validation MSE indicates that the current model captures a substantial portion of variance but leaves room for refinement through advanced feature engineering or the inclusion of interaction terms.

Future work could explore the application of Gradient Boosted Trees, which often outperform Random Forest models on tabular data by sequentially correcting errors. Additionally, a more granular analysis of prediction errors across different price segments could reveal whether the model systematically underperforms on high-value luxury homes compared to mid-market properties.

6 Appendix

Appendix A: Variable Importance Data (Top 10)

The following table details the top ten features ranked by the NUM_AS_ROOT metric extracted from the trained model inspector.

Table 1: Feature importance based on NUM_AS_ROOT Count

Feature Name	NUM_AS_ROOT Count
OverallQual	121
GarageCars	49
ExterQual	40
Neighborhood	35
GrLivArea	21
GarageArea	15
BsmtQual	7
YearBuilt	5
KitchenQual	4
TotalBsmtSF	3

Appendix B: Python Code for Model Training (Excerpt)

The following annotated code snippet illustrates the conversion of the Pandas DataFrame to a TensorFlow Dataset and the compilation of the Random Forest model.

Listing 1: Random Forest Training

```
# Convert training data to tf.data.Dataset format
label = 'SalePrice'
train_ds = tfdf.keras.
    pd_dataframe_to_tf_dataset(
train_ds_pd,
label=label,
task=tfdf.keras.Task.REGRESSION
)

# Instantiate and compile the Random Forest model
rf = tfdf.keras.RandomForestModel(task=tfdf.
    keras.Task.REGRESSION)
rf.compile(metrics=["mse"])

# Train the model
rf.fit(x=train_ds)
```